① Să se studieze dacă următoarele funcții sunt olomorfe sau nu

a) $f(z) = z^2 + i\bar{z}$

$f(x+iy) = (x+iy)^2 + i(x-iy) = x^2 + 2ixy - y^2 + ix + y$

$f = \underbrace{x^2 - y^2 + y}_{P} + i\underbrace{(2xy + x)}_{Q}$

$P, Q \in C^1(\mathbb{R}^2)$

$\frac{\partial P}{\partial x} = 2x$ $\qquad$ $\frac{\partial Q}{\partial y} = 2x$

$\frac{\partial P}{\partial y} = -2y + 1$ $\qquad$ $\frac{\partial Q}{\partial x} = 2y + 1$ $\not\Rightarrow$

$\Rightarrow \begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} \neq -\frac{\partial Q}{\partial x} \end{cases} \Rightarrow$ nu se resp. C-R $\Rightarrow$ f nu e olomorf.

b) $f(z) = e^{2z} + z$

$f(x+iy) = e^{2(x+iy)} + (x+iy) = e^{2x+i2y} + (x+iy)$

$= e^{2x}(\cos 2y + i \sin 2y) + (x+iy)$

$= \underbrace{e^{2x}\cos 2y + x}_{P} + i\underbrace{(e^{2x}\sin 2y + y)}_{Q}$

$\frac{\partial P}{\partial x} = 2 \cdot e^{2x}\cos 2y + 1$ $\qquad$ $\frac{\partial Q}{\partial y} = e^{2x} 2\cos 2y + 1$

$\frac{\partial P}{\partial y} = e^{2x}(-2\sin 2y)$ $\qquad$ $\frac{\partial Q}{\partial x} = 2e^{2x}\sin 2y$ $\not\Rightarrow$

$$\Rightarrow \begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases} \Rightarrow$$ se respectă cond C-R $\Rightarrow$ f olomorfă pe $\mathbb{R}$

② Să se determine punctele în care funcția este olomorfă și să se calculeze derivatele în aceste puncte

$$f(z) = z^2 + 2z\bar{z} - 2\bar{z}^2 + 3z + 2\bar{z}$$

$$z = x + iy$$

$$f(x+iy) = x^2 + 2ixy - y^2 + 2(x^2+y^2) - 2(x^2 - 2ixy - y^2) + 3(x+iy) + 2(x-iy)$$

$$f(x+iy) = x^2 + 2ixy - y^2 + 2x^2 + 2y^2 - 2x^2 + 4ixy + 2y^2 + 3x + 3iy + 2x - 2iy$$

$$f(x+iy) = x^2 + 3y^2 + 5x + 6ixy + iy$$

$$f(x+iy) = \underbrace{x^2 + 5x + 3y^2}_{P} + i\underbrace{(6xy + y)}_{Q}$$

$$\frac{\partial P}{\partial x} = 2x + 5 \qquad \frac{\partial P}{\partial y} = 6y$$

$$\frac{\partial Q}{\partial x} = 6y \qquad \frac{\partial Q}{\partial y} = 6x + 1$$

$$C\text{-}R \quad \begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases} \Rightarrow \begin{cases} 2x+5 = 6x+1 \\ 6y = -6y \end{cases} \Rightarrow \begin{cases} 4x = 4 \\ y = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x = 1 \\ y = 0 \end{cases}$$

$\Rightarrow z = 1 + i \cdot 0 \Rightarrow z = 1 \Rightarrow f$ olomorfă în $z = 1$

$$f'(1) = \frac{1}{2}\left(\frac{\partial f}{\partial x}(1) - i\,\frac{\partial f}{\partial y}(1)\right)$$

$$= \frac{1}{2}\left[\frac{\partial P}{\partial x}(1,0) + i\,\frac{\partial Q}{\partial x}(1,0) - i\left(\frac{\partial P}{\partial y}(1,0) + i\,\frac{\partial Q}{\partial y}(1,0)\right)\right]$$

$$f'(1) = \frac{1}{2}\left[7 + i\cdot 0 - i\left(0 + i\cdot 7\right)\right] = \frac{1}{2}\left(7+7\right) = 7$$

(3) Să se determine $a, b, c \in \mathbb{R}$ a.î.
$f(x+iy) = e^x \cos y + ax + i\left(e^x \sin y + by + cx\right)$ să fie olomorfă

$P, Q \in C^1(\mathbb{R}^2)$

f olomorfă $\Rightarrow$ trebuie să se îndeplinească condițiile $C\text{-}R$

$$\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases} \Rightarrow \begin{cases} e^x \cos y + a = e^x \cos y + b \Rightarrow a = b \\ e^x(-\sin y) = -e^x \sin y + c \Rightarrow c = 0 \end{cases}$$

$\Rightarrow f$ olomorfă $\forall\ a = b \in \mathbb{R}$ și $c = 0$

④ Să se determine funcția olomorfă $f: \mathbb{C} \to \mathbb{C}$ unde $f = P + iQ$, $P(x,y) = e^x(x\cos y - y\sin y)$ și $f(0) = 0$

$$\Delta P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2}$$

$$\frac{\partial P}{\partial x} = e^x(x\cos y - y\sin y) + e^x \cdot \cos y$$

$$\frac{\partial P}{\partial x} = e^x(x\cos y - y\sin y + \cos y)$$

$$\frac{\partial^2 P}{\partial x^2} = e^x(x\cos y - y\sin y + \cos y) + e^x\cos y$$

$$\frac{\partial^2 P}{\partial x^2} = e^x(x\cos y - y\sin y + 2\cos y)$$

$$\frac{\partial P}{\partial y} = e^x(-x\sin y - \sin y - y\cos y)$$

$$\frac{\partial^2 P}{\partial y^2} = e^x(-x\cos y - \cos y - \cos y + y\sin y)$$

$$\frac{\partial^2 P}{\partial y^2} = e^x(-x\cos y + y\sin y - 2\cos y)$$

$$\frac{\partial^2 P}{\partial x^2} = -\frac{\partial^2 P}{\partial y^2} \Rightarrow \Delta P = 0 \Rightarrow P \text{ armonică}$$

cond. C-R
$$\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = e^x(x\cos y - y\sin y + \cos y) \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = e^x(-x\sin y - \sin y - y\cos y) \end{cases}$$

$\xrightarrow[\text{prim.}]{\text{int. pe}}$ $\theta(x,y) = \int e^x(x\cos y - y\sin y + \cos y)\,dy$

$$\theta(x,y) = e^x x\int \cos y\,dy - e^x\int y\sin y\,dy + e^x\int \cos y\,dy$$

$$\int y\sin y\,dy = \int y\cdot(-\cos y)'\,dy = -y\cos y - \int 1(-\cos y)\,dy$$

$$= -y\cos y + \sin y$$

$$\theta(x,y) = e^x x\cdot \sin y - e^x(-y\cos y + \sin y) + e^x \sin y + c(x)$$

$$\theta(x,y) = e^x x\sin y + e^x y\cos y - e^x\sin y + e^x\sin y + c(x)$$

$$\theta(x,y) = e^x(x\sin y + y\cos y) + c(x)$$

$$\frac{\partial\theta}{\partial x} = e^x(x\sin y + \sin y + y\cos y) + c'(x)$$

$$\frac{\partial\theta}{\partial x} = e^x(x\sin y + \sin y + y\cos y)$$

$\Rightarrow c'(x) = 0 \Rightarrow$
$\Rightarrow c(x) = k$
$k \in \mathbb{R}$

$$\theta(x,y) = e^x(x\sin y + y\cos y) + k$$

$f = P + i\theta$

$$f = e^x(x\cos y - y\sin y) + ie^x(x\sin y + y\cos y) + ik$$

$$f = e^x x(\cos y + i\sin y) + e^x y(-\sin y + i\cos y) + ik$$

$$f = e^{x+iy}\cdot x + e^x\cdot y\cdot i(\cos y + i\sin y) + ik$$

$$f = e^{x+iy}x + iye^{x+iy} + ik = e^{x+iy}(x+iy) + ik$$

$f(z) = e^z \cdot z + ik$
$f(0) = 0$
$\Rightarrow ik = 0 \Rightarrow k = 0 \Rightarrow f(z) = e^z \cdot z$

⑤ Să se determine funcția olomorfă $f: \mathbb{C} \to \mathbb{C}$ unde $f = P + iQ$, $P(x,y) = 2\,\mathrm{ch}x \cos y - 3\,\mathrm{ch}x \sin y + x^2 - y^2 + 4xy$

$$\Delta P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2}$$

$$\frac{\partial P}{\partial x} = 2\,\mathrm{ch}x \cos y - 3\,\mathrm{sh}x \sin y + 2x + 4y$$

$$\frac{\partial^2 P}{\partial x^2} = 2\,\mathrm{sh}x \cos y - 3\,\mathrm{ch}x \sin y + 2$$

$$\frac{\partial P}{\partial y} = 2\,\mathrm{sh}x(-\sin y) - 3\,\mathrm{ch}x \cos y - 2y + 4x$$

$$\frac{\partial^2 P}{\partial y^2} = -2\,\mathrm{sh}x \cos y + 3\,\mathrm{ch}x \sin y - 2$$

$$\frac{\partial^2 P}{\partial x^2} = -\frac{\partial^2 P}{\partial y^2} \Rightarrow \Delta P = 0 \Rightarrow P \text{ fc. armonică}$$

cond. C-R
$$\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = 2\,\mathrm{ch}x \cos y - 3\,\mathrm{sh}x \sin y + 2x + 4y \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = 2\,\mathrm{sh}x(-\sin y) - 3\,\mathrm{ch}x \cos y - 2y + 4x \end{cases}$$

$\xrightarrow[\text{prim.}]{\text{int. pe}}$ $Q(x,y) = \int 2\,\mathrm{ch}x \cos y - 3\,\mathrm{sh}x \sin y + 2x + 4y\,dy$

$$Q(x,y) = 2\,\mathrm{ch}x \sin y - 3\,\mathrm{sh}x(-\cos y) + 2xy + 4 \cdot \frac{y^2}{2} + C(x)$$

$$\theta(x,y) = 2\operatorname{ch}x \sin y + 3\operatorname{sh}x \cos y + 2xy + 2y^2 + C(x)$$

$$\xrightarrow[\text{rap. cu } x]{\text{deriv. în}} \frac{\partial \theta}{\partial x} = 2\operatorname{sh}x \sin y + 3\operatorname{ch}x \cos y + 2y + C'(x) \Rightarrow$$

$$\text{f. cond. C-R } \frac{\partial \theta}{\partial x} = 2\operatorname{sh}x \sin y + 3\operatorname{ch}x \cos y + 2y - 4x$$

$$\Rightarrow C'(x) = -4x \Rightarrow C(x) = -4\int x\,dx = -4 \cdot \frac{x^2}{2} + k \,;\ k \in \mathbb{R}$$

$$C(x) = -2x^2 + k \,,\ k \in \mathbb{R}$$

$$\theta(x,y) = 2\operatorname{ch}x \sin y + 3\operatorname{sh}x \cos y + 2xy + 2y^2 - 2x^2 + k$$

$$f = P + i\theta$$

$$f = 2\operatorname{sh}x \cos y - 3\operatorname{ch}x \sin y + x^2 - y^2 + 4xy$$
$$+ i\left(2\operatorname{ch}x \sin y + 3\operatorname{sh}x \cos y + 2xy + 2y^2 - 2x^2 + k\right)$$

$$f = \cancel{2} \cdot \frac{e^x - e^{-x}}{\cancel{2}} \cos y - 3 \cdot \frac{e^x + e^{-x}}{2} \sin y + x^2 - y^2 + 4xy +$$

$$+ i\left(\cancel{2} \cdot \frac{e^x + e^{-x}}{\cancel{2}} \sin y + 3 \cdot \frac{e^x - e^{-x}}{2} \cos y + 2xy + 2y^2 - 2x^2 + k\right)$$

$$f = e^x \overbrace{(\cos y + i \sin y)}^{e^{iy}} - e^{-x} \overbrace{(\cos y - i \sin y)}^{e^{-iy}} + i \cdot \frac{3}{2} e^x (\cos y$$

$$+ i \sin y) - i \cdot \frac{3}{2} e^{-x} (\cos y - i \sin y) + x^2 - y^2 + 2ixy - 2i(x^2 - y^2$$

$$+ 2ixy - 2i(x^2 - y^2 + 2ixy) + ik$$

$$f = e^{x+iy} - e^{-x-iy} + i\cdot\frac{3}{2}e^{x+iy} - i\cdot\frac{3}{2}e^{-x-iy} + (x+iy)^2 - 2i(x+iy)^2 + ik$$

$$f(z) = e^z - e^{-z} + i\cdot\frac{3}{2}e^z - i\cdot\frac{3}{2}e^{-z} + z^2 - 2iz^2 + ik$$

$$f(z) = 2\operatorname{sh} z + 3i\operatorname{sh} z + z^2(1-2i) + ik$$

⑥ Să se determine funcția olomorfă $f: \mathbb{C} \to \mathbb{C}$ unde $f = P + iQ$, $Q(x,y) = \dfrac{\operatorname{sh} 2y}{\cos 2x + \operatorname{ch} 2y}$ și $f\left(\frac{\pi}{4}\right) = 1$

$$\Delta Q = \frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2}$$

$$\frac{\partial Q}{\partial x} = \operatorname{sh} 2y \cdot \frac{0 - 1\cdot(-\sin 2x)\cdot 2}{(\cos 2x + \operatorname{ch} 2y)^2} = \frac{2\operatorname{sh} 2y \sin 2x}{(\cos 2x + \operatorname{ch} 2y)^2}$$

$$(\cos 2x + \operatorname{ch} 2y)(-\sin 2x)\cdot 2$$

$$\frac{\partial^2 Q}{\partial x^2} = 2\cdot \operatorname{sh} 2y \cdot \frac{\cos 2x \cdot 2(\cos 2x + \operatorname{ch} 2y)^2 - \sin 2x \cdot 2}{(\cos 2x + \operatorname{ch} 2y)^{4\,3}}$$

$$\frac{\partial^2 Q}{\partial x^2} = 4\operatorname{sh} 2y \cdot \frac{\cos 2x(\cos 2x + \operatorname{ch} 2y) + 2\sin^2 2x}{(\cos 2x + \operatorname{ch} 2y)^3}$$

$$\frac{\partial^2 Q}{\partial x^2} = 4\operatorname{sh} 2y \cdot \frac{1 + \cos 2x \operatorname{ch} 2y + \sin^2 2x}{(\cos 2x + \operatorname{ch} 2y)^3}$$

$$\frac{\partial Q}{\partial y} = \frac{\operatorname{ch} 2y \cdot 2(\cos 2x + \operatorname{ch} 2y) - \operatorname{sh} 2y \cdot \operatorname{sh} 2y \cdot 2}{(\cos 2x + \operatorname{ch} 2y)^2}$$

$$\frac{\partial Q}{\partial y} = 2\,\frac{\cos 2x \operatorname{ch} 2y + \operatorname{ch}^2 2y - \operatorname{sh}^2 2y}{(\cos 2x + \operatorname{ch} 2y)^2}$$

$$\operatorname{ch} 2y = \frac{e^{2y} + e^{-2y}}{2} \Rightarrow \operatorname{ch}^2 2y = \frac{e^{4y} + 2 + e^{-4y}}{4}$$

$$\operatorname{sh}^2 2y = \frac{e^{4y} - 2 + e^{-4y}}{4}$$

$$\frac{\partial Q}{\partial y} = 2 \cdot \frac{\cos 2x \operatorname{ch} 2y + 1}{(\cos 2x + \operatorname{ch} 2y)^2}$$

$$\frac{\partial^2 Q}{\partial y^2} = 2 \cdot \frac{\cos 2x \operatorname{sh} 2y \cdot 2(\cos 2x + \operatorname{ch} 2y)^2 - (\cos 2x \operatorname{ch} 2y + 1) \cdot 2(\cos 2x + \operatorname{ch} 2y) \cdot \operatorname{sh} 2y \cdot 2}{(\cos 2x + \operatorname{ch} 2y)^4}$$

$$\frac{\partial^2 Q}{\partial y^2} = 4 \operatorname{sh} 2y \, \frac{\cos 2x (\cos 2x + \operatorname{ch} 2y) - 2(\cos 2x \operatorname{ch} 2y + 1)}{(\cos 2x + \operatorname{ch} 2y)^3}$$

$$\frac{\partial^2 Q}{\partial y^2} = 4 \operatorname{sh} 2y \, \frac{\cos^2 2x + \cos 2x \operatorname{ch} 2y - 2\cos 2x \operatorname{ch} 2y - 2}{(\cos 2x + \operatorname{ch} 2y)^3}$$

$$\frac{\partial^2 Q}{\partial y^2} = 4 \operatorname{sh} 2y \, \frac{\cos^2 2x - \cos 2x \operatorname{ch} 2y - 2}{(\cos 2x + \operatorname{ch} 2y)^3}$$

$$\Delta Q = 4 \operatorname{sh} 2y \cdot \left(\frac{1 + \cos 2x \operatorname{ch} 2y + \sin^2 2x}{(\cos 2x + \operatorname{ch} 2y)^3} + \frac{\cos^2 2x - \cos 2x \operatorname{ch} 2y - 2}{(\cos 2x + \operatorname{ch} 2y)^3} \right)$$

$\Rightarrow \Delta Q = 0 \Rightarrow Q$ fc. armonică

$$C\text{-}R \Rightarrow \begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = 2 \cdot \frac{\cos 2x \operatorname{ch} 2y + 1}{(\cos 2x + \operatorname{ch} 2y)^2} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = -\frac{2 \operatorname{sh} 2y \sin 2x}{(\cos 2x + \operatorname{ch} 2y)^2} \xrightarrow{\text{integ. prin } y} \end{cases}$$

$$P(x,y) = -\sin 2x \int \frac{2 \operatorname{sh} 2y}{(\cos 2x + \operatorname{ch} 2y)^2} dy$$

Notăm $t = \operatorname{ch} 2y$

$dt = 2 \operatorname{sh} 2y \, dy$

$$P(x,y) = -\sin 2x \int \frac{1}{(\cos 2x + t)^2} dt = -\sin 2x \cdot \frac{-1}{\cos 2x + t} + C(x)$$

$$P(x,y) = \frac{\sin 2x}{\cos 2x + \operatorname{ch} 2y} + C(x)$$

$$\frac{\partial P}{\partial x} = \frac{2 \cos 2x (\cos 2x + \operatorname{ch} 2y) - \sin 2x \cdot (-2 \sin 2x)}{(\cos 2x + \operatorname{ch} 2y)^2} + C'(x)$$

$\Rightarrow$

$$\frac{\partial P}{\partial x} = 2 \cdot \frac{\cos 2x \operatorname{ch} 2y + 1}{(\cos 2x + \operatorname{ch} 2y)^2}$$

$\Rightarrow C'(x) = 0 \Rightarrow C(x) = k, \; k \in \mathbb{R}$

$f = P + iQ$

$$f = \frac{\sin 2x}{\cos 2x + \operatorname{ch} 2y} + k + i \frac{\operatorname{sh} 2y}{\cos 2x + \operatorname{ch} 2y} \qquad k \in \mathbb{R}$$

$$f = \frac{\sin 2x + i \operatorname{sh} 2y}{\cos 2x + \operatorname{ch} 2y} + k \qquad k \in \mathbb{R}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\sin(2iy) = \frac{e^{-2y} - e^{2y}}{2i} = -\frac{1}{i} \operatorname{sh} 2y = i \operatorname{sh} 2y$$

$$\cos(2iy) = \frac{e^{-2y} + e^{2y}}{2} = \operatorname{ch} 2y$$

$$f = \frac{\sin 2x + \sin(2iy)}{\cos 2x + \cos(2iy)} + k = \frac{2 \sin(x+iy)\cos(x-iy)}{2 \cos(x+iy)\cos(x-iy)} + k$$

$$f(z) = \operatorname{tg} z + k$$

$$f(\pi/4) = 1 \Rightarrow k = 0 \Rightarrow f(z) = \operatorname{tg} z$$